Problem Set 1 — Likelihood

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Covers sessions 1–4 (prerequisites, statistical models, maximum likelihood, likelihood inference, numerical optimisation; Labs 00–01). Show derivations and interpret every number in context. Notation: $\ell(\theta) = \log L(\theta)$ is the log-likelihood, $S(\theta) = \ell'(\theta)$ the score, $I(\theta) = -\ell''(\theta)$ the *observed* information of the whole sample, $\mathcal{I}(\theta) = \mathbb{E}\{I(\theta)\}$ the *expected* (Fisher) information, and $\mathrm{se}(\hat\theta) = I(\hat\theta)^{-1/2}$. Unless a problem says otherwise, use $\alpha = 0.05$. Code that you need to read is shown; the rest you write yourself. Set a seed whenever you simulate. Allowed tools: R, a calculator; GenAI may be used for coding help with acknowledgement but not to produce your answers.

1 Prerequisites: distributions, matrices and iterated expectations

(a) Let $Y \sim \mathrm{Gamma}(a, b)$ with density $f(y) = b^a y^{a-1} e^{-by}/\Gamma(a)$, $y > 0$ (shape a, rate b). Using only the definition $\Gamma(a) = \int_0^\infty u^{a-1}e^{-u}du$ and $\Gamma(a+1) = a\Gamma(a)$, show that $\mathbb{E}(Y) = a/b$ and $\mathrm{Var}(Y) = a/b^2$.

(b) Let $Y_1, \ldots, Y_n$ be iid $\mathrm{Exp}(\lambda)$ (rate λ). Find the distribution of $\min_i Y_i$ and of $T = \sum_i Y_i$.

(c) Suppose $Y \mid \Lambda \sim \mathrm{Poisson}(\Lambda)$ and $\Lambda \sim \mathrm{Gamma}(a, b)$. Use the laws of total expectation and total variance to find $\mathbb{E}(Y)$ and $\mathrm{Var}(Y)$. Is Y over- or under-dispersed relative to a Poisson, and why does this matter for the model in Problem 5? Check your formulas by simulation with $a = 3$, $b = 0.5$ and 10^6 draws.

(d) A small regression has design matrix X with intercept column and $x = (1, 2, 4, 5)^\top$. Compute $X^\top X$, its determinant and inverse by hand, and its eigenvalues (R is allowed for the eigenvalues; check that the product of the eigenvalues equals the determinant). Is $X^\top X$ positive definite? What is its condition number $\lambda_{\max}/\lambda_{\min}$ and what would a very large value indicate?

2 Statistical models as approximations: the Nile

The series `Nile` contains the annual flow (in $10^8\,\mathrm{m}^3$) of the river Nile at Aswan, 1871–1970 ($n = 100$). A first analyst proposes the model $y_i \overset{iid}{\sim} N(\mu, \sigma^2)$.

(a) Write down $\ell(\mu, \sigma^2)$, find the MLEs, and show that the observed information at the MLE is $\mathrm{diag}\{n/\hat\sigma^2,\ n/(2\hat\sigma^4)\}$. Compute $\hat\mu$, $\hat\sigma^2$ and $\mathrm{se}(\hat\mu)$ for the Nile data. Is $\hat\sigma^2$ unbiased? Is it consistent?

(b) Plot the series and its autocorrelation function (`acf`). Which assumption of the model does the plot call into question, and what does the lag-1 autocorrelation r_1 tell you? For a stationary AR(1)-type series the variance of $\bar y$ is approximately inflated by the factor $(1+\rho)/(1-\rho)$ relative to the iid case. Use r_1 for ρ to compute an effective sample size and the corrected standard error of $\hat\mu$.

(c) A second analyst notices that the Aswan dam started operating in 1899 and fits two means (1871–1898 and 1899–1970) with a common variance. Compute the maximised log-likelihood and the AIC of both models. Why is the LR statistic $2\{\ell_1 - \ell_0\}$ not an honest χ^2_1 test here? Do the residuals of the second model look more like iid noise?

(d) *Communicate.* In about 150 words, explain to a non-statistician why the first model, although a perfectly legitimate likelihood model, gives an overconfident statement about the average flow, and what the second analysis suggests. Mention one limitation of the second.

3 The Pareto model for large rivers

The `rivers` data give the lengths (miles) of 141 major North American rivers. A Pareto model with *known* threshold $x_m > 0$ has density $f(y;\theta) = \theta x_m^\theta y^{-(\theta+1)}$ for $y \geq x_m$, with $\theta > 0$.

(a) For an iid sample of size n above x_m, write $\ell(\theta)$, $S(\theta)$ and $I(\theta)$, and find $\hat\theta$. Show that $I(\theta)$ does not depend on the data, so that $I(\theta) = \mathcal{I}(\theta)$ for all θ and not only at $\hat\theta$.

(b) Show that $\log(Y/x_m) \sim \text{Exp}(\theta)$, hence $T = \sum_i \log(y_i/x_m) \sim \text{Gamma}(n,\theta)$. Use $\mathbb{E}(1/T) = \theta/(n-1)$ (which follows from the Gamma density) to show that $\hat\theta$ is biased, and give an unbiased estimator. Quantify the bias for $n = 77$.

(c) The tail of the distribution is the interesting part. Use $x_m = 400$ and the $n = 77$ rivers with length ≥ 400 miles. Compute $\hat\theta$, $\text{se}(\hat\theta)$ and a 95% Wald interval. For $\theta \leq 1$ the Pareto mean is infinite; is that a concern for these data?

(d) By the invariance property, give the MLE of $q = P(Y > 800 \mid Y \geq 400) = (400/800)^\theta$ and its standard error by the delta method. Compare with the observed proportion among the 77 rivers.

(e) Check the model: make a plot of $\log(y/x_m)$ against exponential quantiles, and compare the Pareto fit with $x_m = 135$ (whole sample, $n = 141$) to the data. Which is the better use of the Pareto model, and what does this say about "all models are wrong"?

4 Censored survival times

In the `aml` data of the `survival` package, patients with acute myelogenous leukaemia in remission were followed for the time (weeks) to relapse; some patients are lost to follow-up (`status = 0`, right censored). For the *maintained* chemotherapy group ($n = 11$) the times are $9, 13, 13^+, 18, 23, 28^+, 31, 34, 45^+, 48, 161^+$ (the $^+$ marks censoring). Assume $T_i \sim \text{Exp}(\lambda)$ (rate λ) with non-informative censoring.

(a) An event contributes $f(t_i) = \lambda e^{-\lambda t_i}$ and a censored observation contributes $S(t_i) = e^{-\lambda t_i}$ to the likelihood. Show that $\ell(\lambda) = d\log\lambda - \lambda\sum_i t_i$ where d is the number of events, and find $\hat\lambda$, $I(\lambda)$ and $\text{se}(\hat\lambda)$. Compute them for the maintained group.

(b) Compute the median survival time $\log 2/\hat\lambda$ and a 95% interval for it, in two ways: from a Wald interval for λ and from the likelihood-ratio interval for λ. Which would you prefer with $d = 7$ events?

(c) A colleague "ignores" censoring by treating all 11 times as relapse times. Compute her estimate of the mean time to relapse and compare with yours. In which direction is she biased, and why?

(d) The non-maintained group has $d = 11$ events in 255 patient-weeks. Test $H_0 : \lambda_1 = \lambda_2$ with a likelihood-ratio test (derive the pooled MLE). Report the statistic, p-value and a one-sentence conclusion. Which assumption of the exponential model would you check next, and how?

5 Likelihood inference for a small binomial sample

In a quality audit, $y = 2$ of $n = 25$ randomly sampled invoices contain an error. Model $y \sim \text{Bin}(n,\theta)$.

(a) Derive $\hat\theta$, $I(\theta)$, $\mathcal{I}(\theta)$, and show that $I(\hat\theta) = \mathcal{I}(\hat\theta)$ but $I(\theta) \neq \mathcal{I}(\theta)$ for $\theta \neq \hat\theta$.

(b) Compute the 95% Wald interval for θ and comment. Then compute the 95% likelihood-ratio interval (numerically in R, with `uniroot`) and the score (Wilson) interval $\{\theta : (\hat\theta - \theta)^2/\{\theta(1-\theta)/n\} \leq 3.84\}$. Plot $\ell(\theta)$ with the three intervals.

(c) Reparametrise to the log-odds $\psi = \log\{\theta/(1-\theta)\}$. Derive $\text{se}(\hat\psi)$ by the delta method, form a Wald interval for ψ and back-transform it. What does this do relative to (b)? What would the LR interval for ψ be, after back-transforming?

(d) Test $H_0 : \theta = 0.20$ (the error rate the company claims) with the Wald, score and LR statistics. Give p-values. Why do the three disagree here, and which do you trust?

(e) Simulation: with $\theta = 0.08$ and $n = 25$, estimate by Monte Carlo (10^4 replicates, set a seed) the actual coverage of the nominal-95% Wald, LR and score intervals. For the LR interval it is enough to check, for each replicate, whether $2\{\ell(\hat\theta) - \ell(0.08)\} \leq 3.84$ (use $0\log 0 = 0$). Explain what you find, in particular the role of the samples with $y = 0$.

6 A two-parameter model: rainfall and the gamma distribution

The `precip` data give the average annual precipitation (inches) in $n = 70$ US cities. Consider $y_i \overset{iid}{\sim} \text{Gamma}(\alpha, \beta)$ with shape α and rate β.

(a) Write $\ell(\alpha, \beta)$. For fixed α show that $\hat\beta(\alpha) = \alpha/\bar y$ and obtain the profile log-likelihood $\ell_p(\alpha) = n\{\alpha \log(\alpha/\bar y) - \log\Gamma(\alpha) + (\alpha - 1)\overline{\log y} - \alpha\}$.

(b) Maximise ℓ_p numerically (`optimize` or `optim`) to get $\hat\alpha$ and $\hat\beta$, and obtain a 95% likelihood interval for α by solving $2\{\ell_p(\hat\alpha) - \ell_p(\alpha)\} = \chi^2_{1,0.95}$. Test $H_0 : \alpha = 1$ (exponential) with an LR test. Interpret.

(c) Reparametrise by (μ, α) with mean $\mu = \alpha/\beta$. Show that the mixed second derivative $\partial^2\ell/\partial\mu\,\partial\alpha$ is zero at $\mu = \bar y$ (the parameters are *orthogonal*), that $\hat\mu = \bar y$ and $I_{\mu\mu}(\hat\mu) = n\hat\alpha/\bar y^2$. Hence give $\text{se}(\hat\mu)$ and a 95% interval for the mean rainfall. Compare with $s/\sqrt{n}$.

(d) Fit the model also with `optim(..., method = "BFGS", hessian = TRUE)` on the scale $(\log\alpha, \log\beta)$. Report $\hat\alpha$ and the standard error of $\log\hat\alpha$ from `solve(hessian)`, and use it to build a Wald interval for α. Compare with (b). Why is optimising on the log scale a good idea?

(e) Fit the lognormal model to the same data (closed-form MLEs). Compare the maximised log-likelihoods and AICs. The two models are not nested: why can you still compare AIC but not use an LR χ^2 test?

7 Numerical optimisation: the Cauchy location model

Seven repeated readings of a quantity from a heavy-tailed instrument are $y = (-4.2,\ -0.9,\ 0.4,\ 1.1,\ 1.8,\ 7.9,\ 8.6)$. Assume $y_i \overset{iid}{\sim} \text{Cauchy}(\mu, 1)$, with density $f(y) = 1/[\pi\{1 + (y - \mu)^2\}]$.

(a) Derive $S(\mu)$ and $I(\mu)$. Show that the expected information is $\mathcal{I}(\mu) = n/2$, using $\mathbb{E}\{(1 - Z^2)/(1 + Z^2)^2\} = 1/4$ for $Z = Y - \mu$ (you may take this integral as given, but state why $\mathcal{I}$ is free of μ).

(b) Plot $\ell(\mu)$ for $\mu \in (-8, 14)$. How many local maxima does it have? Find them with `optim` from several starting values. Which is the MLE?

(c) Implement Newton–Raphson $\mu_{k+1} = \mu_k + S(\mu_k)/I(\mu_k)$ and Fisher scoring $\mu_{k+1} = \mu_k + S(\mu_k)/\mathcal{I}(\mu_k)$. Start both from $\mu_0 \in \{\bar y,\ \text{median}(y),\ 5,\ 8\}$ for 20 iterations and tabulate where they end. Explain what happens from $\mu_0 = 5$: examine I at the limit of Newton–Raphson.

(d) At the MLE, compute the standard error using the observed information and using the expected information. Which would you report, and why? How different are they for $n = 7$?

(e) *Critique.* A colleague says: "The sample mean is the MLE of a location parameter, so I'll use $\bar y = 2.1$." Explain in two or three sentences why this is wrong for the Cauchy model.

8 Critique: true or false?

(a) Each statement below was written by a student. Say whether it is correct, and if not, correct it in one or two sentences (use a counter-example or a formula).

1. "$L(\theta)$ is the probability that the parameter equals θ, so $\int L(\theta)\,d\theta = 1$."
2. "The MLE $\hat\sigma^2 = \frac{1}{n}\sum(y_i - \bar y)^2$ is biased, so it is a bad estimator and $s^2 = \frac{1}{n-1}\sum(y_i - \bar y)^2$ should always be preferred."
3. "Wald tests are invariant to reparametrisation, the LR test is not."
4. "For any model, the standard error of the MLE is $\sigma/\sqrt{n}$."
5. "If the log-likelihood has a stationary point, Newton–Raphson will find the MLE."
6. "The 95% LR interval is $\{\theta : \ell(\theta) \geq \ell(\hat\theta) - 1.96\}$."

(b) Check statement 2 by simulation: for $n = 10$, $\sigma^2 = 1$, estimate by Monte Carlo (10^5 replicates, with a seed) the bias and the mean squared error of both estimators, and compare with the exact values $\mathbb{E}(\hat\sigma^2) = (n-1)/n$, $\text{MSE}(\hat\sigma^2) = (2n-1)/n^2$, $\text{MSE}(s^2) = 2/(n-1)$.

(c) *Communicate.* In about 150 words, write for a non-statistician manager the conclusion of Problem 5: what can be said about the invoice error rate from 2 errors in 25 invoices, and why the textbook formula "estimate ± 2 standard errors" should not be used here.